

LegalLens – AI-Powered Multi-Agent Legal Document Analyser

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Introduction — Why this matters

- Everyday contracts like rental agreements, job contracts, loans, and NDAs are often difficult to understand.
- The legal language is complicated, so many people misunderstand their rights, responsibilities, and risks.
- Hiring a lawyer to review a contract can be expensive and take a long time.
- AI can help by explaining contracts in simple language, pointing out important risks, and suggesting ways to negotiate better terms.
- **LegalLens** helps people understand contracts quickly, affordably, and through an easy-to-use interactive experience.



Purpose

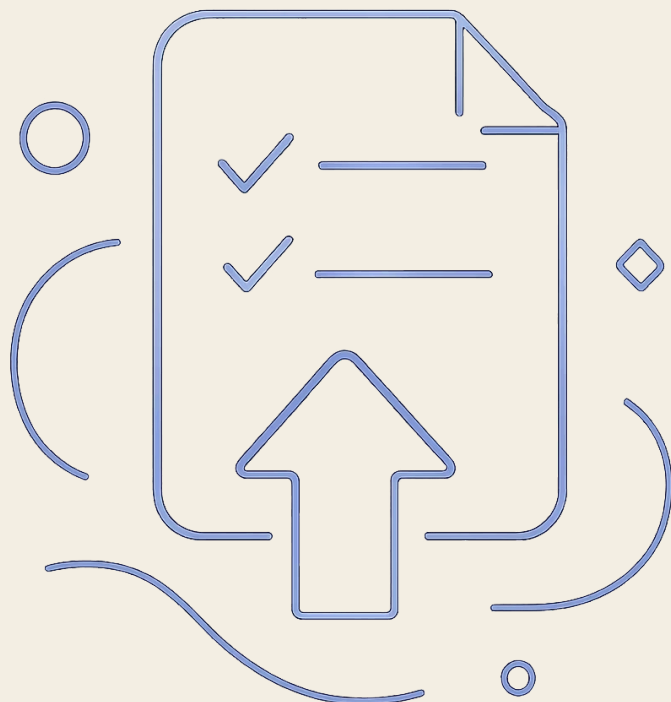
- Simplify legal language into clear, plain English explanations.
- Automatically detect and flag risky clauses and hidden obligations.
- Suggest negotiable alternatives and standard-safe phrasing.
- Provide concise summaries and structured reports for quick decisions.
- Offer an interactive assistant to ask questions about the document.

Need for the Proposed System

- **Fast:** Quickly analyzes contract clauses and explains them in simple English.
- **Affordable:** Helps reduce the need for expensive legal consultations for basic contract reviews.
- **Useful:** Highlights important risks and suggests changes you can negotiate.
- **Interactive:** Lets you ask questions about your contract through a chat-like assistant.
- **Easy to Review:** Generates clear reports and structured data that can be shared, reviewed, or used for compliance.

Quick Comparison

Existing System	Proposed (LegalLens)
Made for companies	Made for everyone
Complex legal reports	Simple English summaries
Basic risk detection	Risk + negotiation suggestions
Separate review process	All-in-one contract analysis
Focus on contract management	Focus on contract understanding



Objectives

- Allow users to upload PDFs and extract full text reliably.
- Automatically identify and extract clauses and clause types.
- Detect risky terms and assign risk categories (high/medium/low).
- Generate plain-English explanations and negotiation suggestions.
- Produce structured summaries and exportable reports.
- Provide a chat interface for question-driven exploration.

Key Features & Modules

1. User Authentication & Profile

Manages secure user registration, login, password reset, and JWT-based authentication. Ensures only authorized users can access the platform..

2. Document Upload & Processing

Allows users to securely upload legal documents in PDF format. The system validates the file, extracts the document text, and prepares it for AI-based clause analysis and processing.

3. Clause Detection & Classification

Identifies and extracts important legal clauses from the uploaded document using AI. The system classifies clauses into categories such as payment, termination, confidentiality, liability, and renewal for further analysis.

4. Proactive Risk & Obligation Detection

Evaluates extracted clauses to identify potential legal risks, hidden conditions, and unfavorable terms. Assigns a risk level to each clause..

Key Features & Modules (Continued)

5. Clause Simplification Module

Converts complex legal language into simple, easy-to-understand explanations so non-legal users can understand contracts.

6. Contextual Negotiation Guidance

Recommends clauses that are commonly negotiable and suggests alternative wording or negotiation strategies.

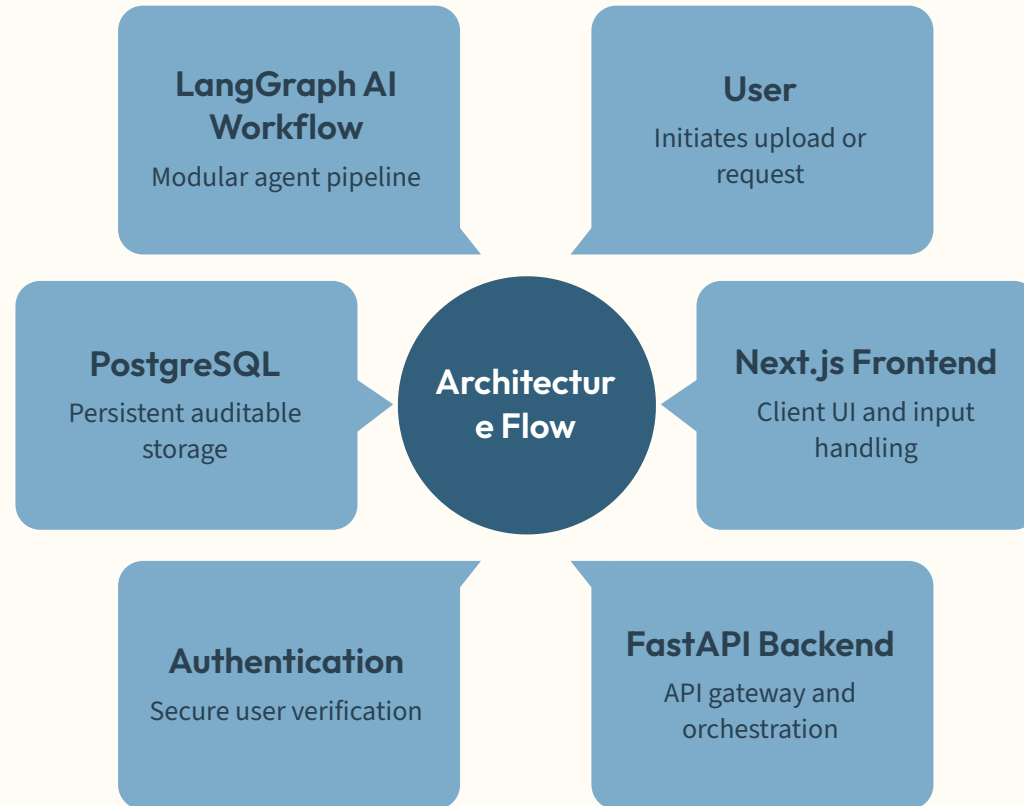
7. AI Report Generator

Compiles analysis results into a structured report containing clause summaries, risk assessment, simplified explanations, and negotiation suggestions.

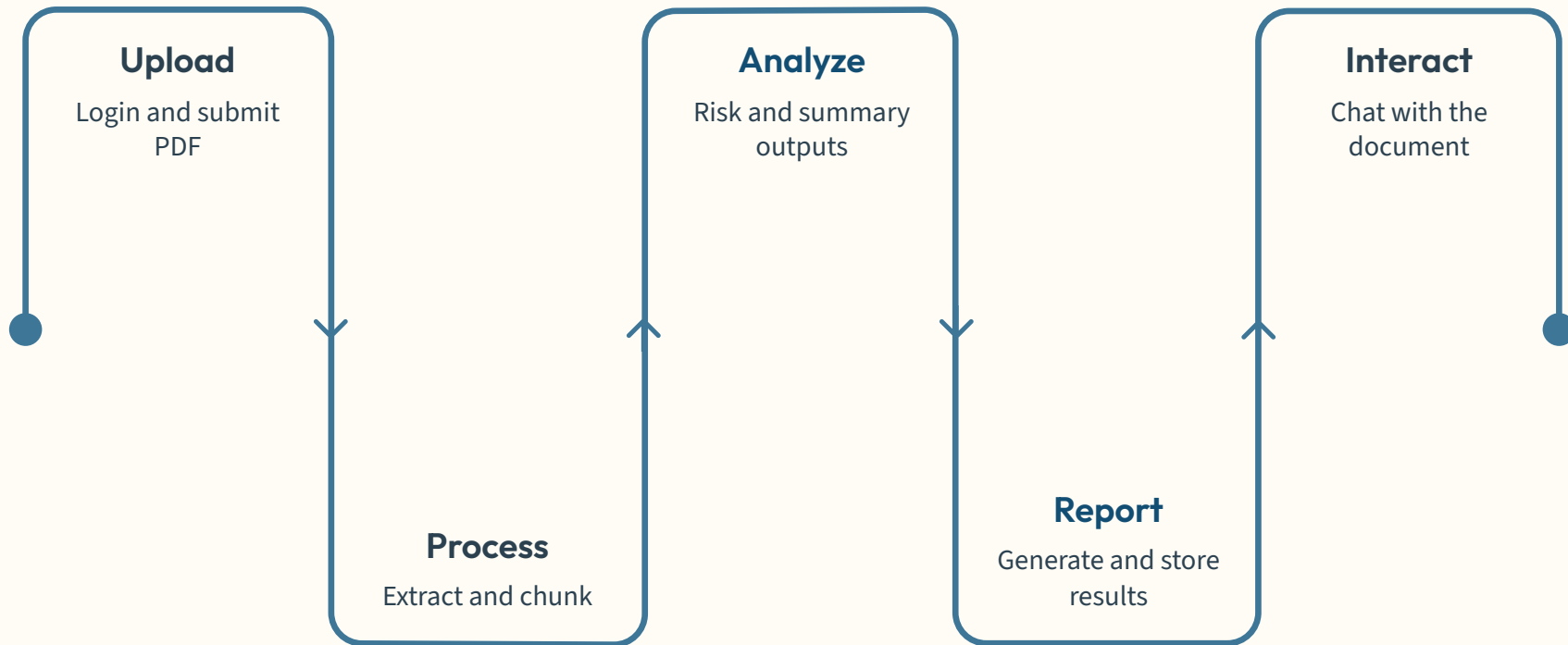
8. Interactive Q&A Chat Agent

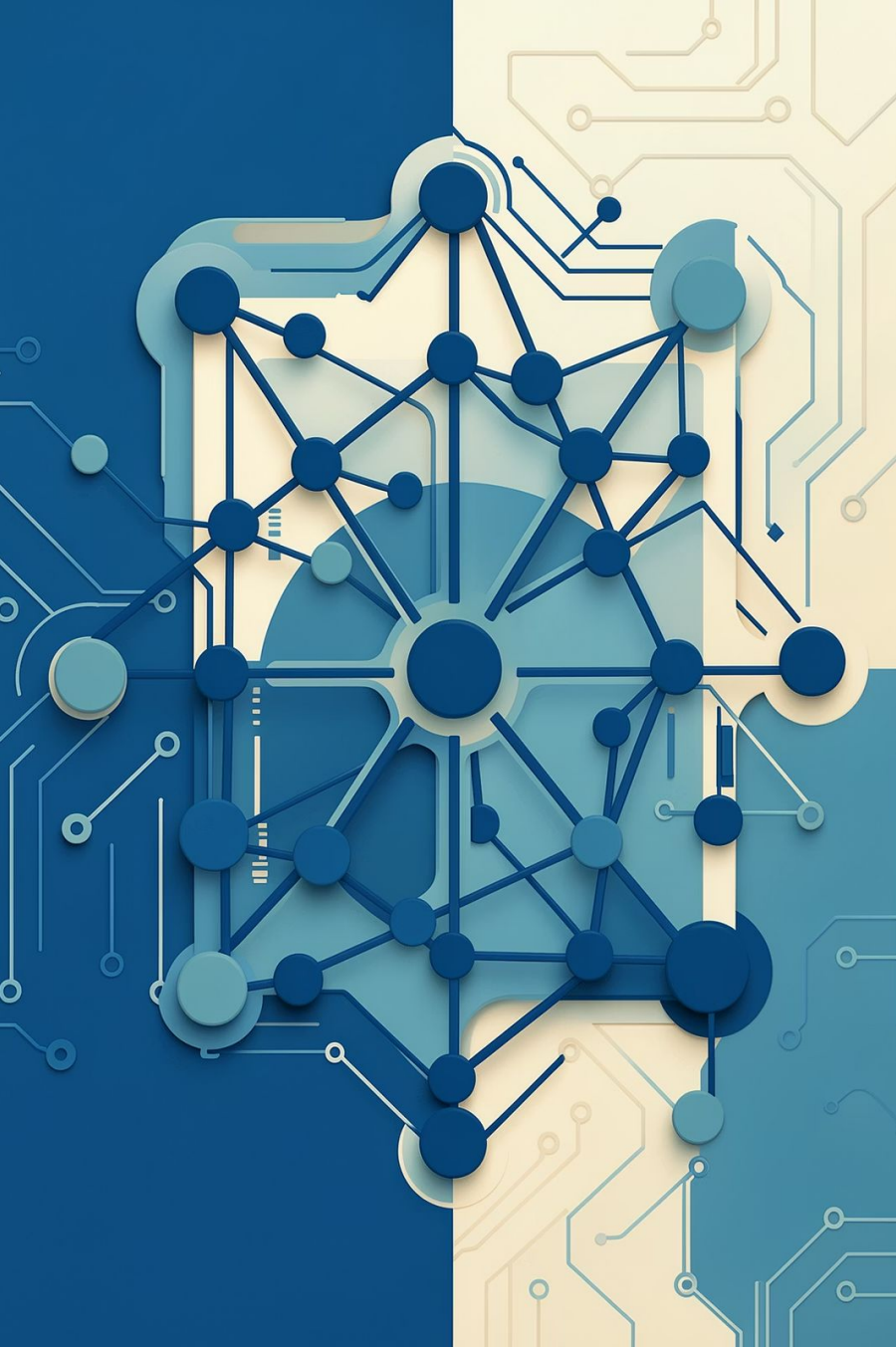
Enables users to ask questions about the uploaded document and receive context-aware answers using Retrieval-Augmented Generation (RAG).

System Architecture



Workflow Diagram





Key Technologies

Application Stack

- Next.js (Frontend)
- FastAPI (Backend API)
- LangGraph (AI Orchestration)
- ProgressSQL(Database)

AI & Data Processing

- Large Language Models (LLMs)
- Vector Databases (RAG)
- PDF Parsers & OCR Libraries



Thank You!